CLOXACILLIN

Empty stomach (PO)

Penicillin allergy

CHEMICAL NAME (2S,5R,6R)-6-[[3-(2-chlorophenyl)-5-methyl-1,2-oxazole-4-carbonyl]amino]-3,3-dimethyl-7-oxo-4-thia-1-azabicyclo[3.2.0]heptane-2-carboxylic acid (sodium)

CLASS Penicillinase-resistant (anti-staphylococcal) penicillin

TRADE NAMES Orbenin · Cloxapen · Tegopen

THERAPEUTIC USES / INDICATIONS

- Infections due to penicillinase-producing *Staph. aureus* (MSSA): skin and soft tissue, impetigo, cellulitis.
- Osteomyelitis, septic arthritis.
- Staphylococcal pneumonia, endocarditis (IV).

PHARMACODYNAMICS

- Side chain protects β-lactam ring from **staphylococcal penicillinase** → active against MSSA.
- Inhibits cell-wall synthesis → bactericidal.
- Not active against MRSA.

ADMINISTRATION

- **PO:** 1 h before or 2 h after food.
- **IV:** slow injection over 3–4 min or infusion; **IM**.
- Often combined with ampicillin (ampiclox) in some regions.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, diarrhoea.
- Rash.
- Phlebitis (IV).

ADVERSE EFFECTS (SERIOUS)

- Anaphylaxis.
- Cholestatic hepatitis.
- Neutropenia with prolonged use.
- *C. difficile* colitis.

DOSAGES

- **Adult PO:** 500 mg every 6 h.
- **Adult IV/IM:** 500 mg–2 g every 6 h (severe up to 12 g/day).
- **Children:** 50–100 mg/kg/day divided every 6 h (higher in severe infection).

CONTRAINDICATIONS & CAUTIONS

- Penicillin hypersensitivity.
- **Caution:** hepatic impairment, neonates (jaundice).

KEY INTERACTIONS

- **Methotrexate:** ↑ toxicity.
- **Warfarin:** altered INR.

PHARMACOKINETICS

- **Peak:** PO 1 h. Food reduces absorption.
- Highly protein-bound; some liver metabolism.
- Excreted in urine and bile. **t½:** 30–60 min.

NURSING MANAGEMENT

- Check allergy history.
- Give PO on an empty stomach.
- Monitor LFTs and CBC in long courses (osteomyelitis).
- Monitor IV site and wound healing.
- Track cultures (MRSA needs vancomycin).

PATIENT EDUCATION

- Take tablets on an empty stomach.
- Finish the course; bone infections need long treatment.
- Report yellow skin or severe diarrhoea.

KEY POINTS

- Anti-staphylococcal penicillin (MSSA).
- Empty stomach orally.
- Ineffective against MRSA.
- Watch LFTs and CBC in long courses.